



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to KAMATCHIAMMAN ENTERPRISES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

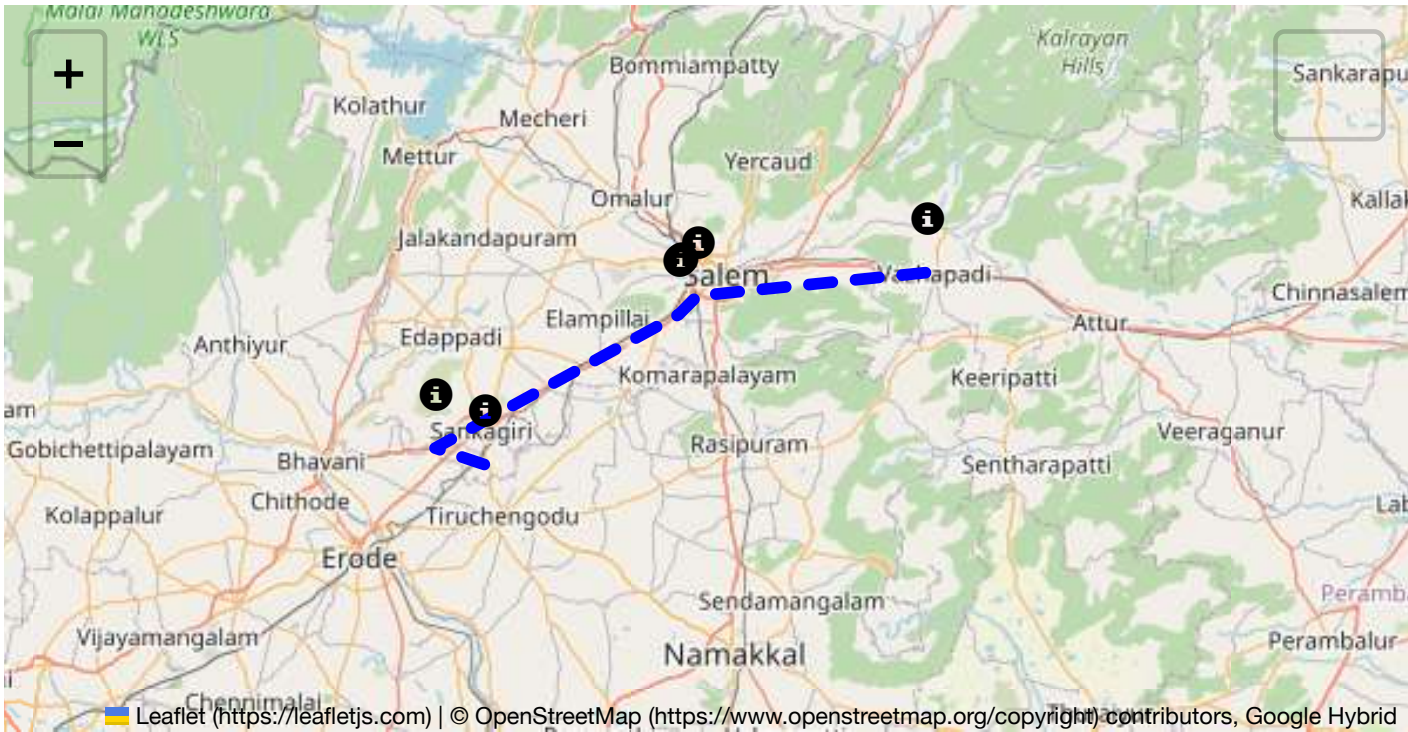
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.

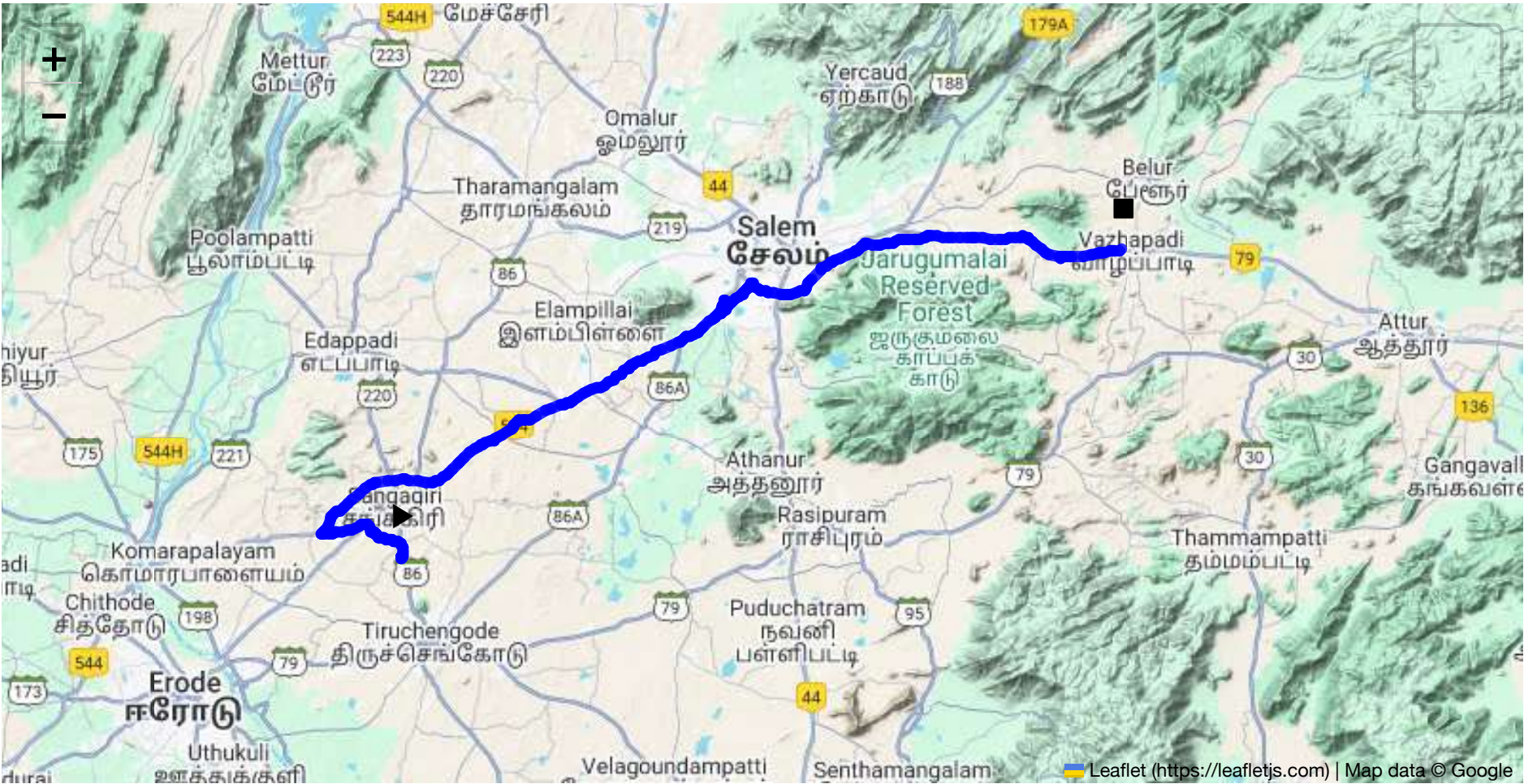


Route Summary:
Total Distance: 83.01 km
Estimated Duration: 1.6 hours

Adjusted Duration (Heavy Vehicle): 2.0 hours

Start: (11.4381, 77.8734)

End: (11.656943, 78.3937)



Welcome to the Journey Risk Management Study

Certainly! Here is a comprehensive analysis of the route from Sangagiri to Vazhapadi, Tamil Nadu, incorporating all the specified factors:

1. Overview of the Route Map

The route travels approximately 83 kilometers, mostly through urban and semi-urban areas. It begins in Sangagiri, passes through Bhavani Main Road, Uthamasolapuram, and Kondalampatti before reaching Vazhapadi. The route primarily follows major roads and highways that are well-trafficked.

2. Typical Weather Conditions and Potential Weather-Related Hazards

- **Climate:** Tropical with hot and dry summers, and significant rainfall during the monsoon season (June to September).
- **Hazards:** Monsoon rains can cause flooding and road damage, while summers may result in heat stress for vehicles and drivers. Fog is rare but possible during cooler months.

3. Analysis of Traffic Patterns

- **Peak Hours:** Congestion is prevalent during the morning rush (8:00 AM - 10:00 AM) and evening (5:00 PM - 7:00 PM), particularly near urban centers like Salem.
- **Congestion-Prone Areas:** Salem city and its suburbs often experience slow traffic; vigilance is necessary in these areas.

4. Assessment of Road Quality and Infrastructure

- **General:** Roads are generally paved and in decent condition, but with varying maintenance levels.
- **Concerns:** Potholes and uneven surfaces may be present, particularly after monsoon rains. The infrastructure in rural segments may not always support heavy vehicles comfortably.

5. Suggestions for Alternative Routes for Emergencies

- **From Sangagiri to Salem:** Consider bypassing Salem via SH 86 which might provide a faster alternative during city congestion.
- **Local bypasses:** Utilize local roads if major roads are blocked, referencing local maps and GPS.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- **Permits:** Required for transporting hazardous materials, secured well in advance.
- **Time constraints:** Movement during certain high-traffic periods may be restricted.
- **Speed limits:** Strict adherence needed, particularly in urban areas.

7. Overview of Historical Incidents

- There have been a few incidents involving heavy vehicles; most involve vehicular malfunction or road conditions post-rains. Staying informed about past patterns can help anticipate potential risks.

8. Environmental Considerations and Sensitive Areas

- **Water Bodies:** Proximity to water bodies requires extra caution to prevent contamination.
- **Wildlife:** Rural stretches may be near protected areas; emergency plans should minimize environmental impacts.

9. Analysis of Communication Coverage

- **Coverage:** Generally good along the route in urban areas. Potential signal loss may occur in rural segments; ensure all communication devices are equipped for such scenarios.

10. Estimated Emergency Response Times

- **Sangagiri to Salem:** Approx. 30-45 minutes depending on conditions.
- **Salem to Vazhapadi:** Slower response, averaging 30-60 minutes due to more rural and less accessible locations.

12. Overall Summary of Risk Assessment

- **Moderate Risk:** The route is well-frequented but requires vigilance due to potential congestion, varying road quality, and weather impacts.
- **Mitigation Measures:** Preparing for diverse weather conditions, securing proper permits, planning around traffic peaks, and staying updated on road and communication conditions will help in maintaining safety.

In structuring this route plan, it is important that truck drivers are thoroughly briefed on these points to ensure safety and efficiency in their transport operations. Equipment checks, route flexibility, and informed planning are vital for reducing risks.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63266, 78.12699	15 KM/Hr
1	Turn	Medium	11.43822, 77.87348	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	Medium	11.44900, 77.87406	30 KM/Hr
5	Turn	Medium	11.45358, 77.85781	30 KM/Hr
6	Turn	High	11.45353, 77.85697	15 KM/Hr
7	Turn	Medium	11.45853, 77.85139	30 KM/Hr
8	Turn	Medium	11.45906, 77.85132	30 KM/Hr
9	Turn	Medium	11.46307, 77.84941	30 KM/Hr
10	Turn	Medium	11.46316, 77.84921	30 KM/Hr
11	Turn	High	11.45542, 77.81725	15 KM/Hr
12	Turn	High	11.45636, 77.81678	15 KM/Hr
13	Blind Spot	Blind Spot	11.62145, 78.10740	10 KM/Hr
14	Turn	High	11.62130, 78.10776	15 KM/Hr
15	Turn	High	11.61235, 78.10438	15 KM/Hr
16	Turn	High	11.61219, 78.10450	15 KM/Hr
17	Turn	Medium	11.63018, 78.12472	30 KM/Hr
18	Blind Spot	Blind Spot	11.63028, 78.12474	10 KM/Hr
19	Blind Spot	Blind Spot	11.62986, 78.12425	10 KM/Hr
20	Turn	High	11.63304, 78.12738	15 KM/Hr
21	Turn	High	11.65458, 78.37988	15 KM/Hr
22	Turn	High	11.65479, 78.37995	15 KM/Hr
23	Turn	Medium	11.65602, 78.38701	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
4	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
5	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
6	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
9	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
13	hospital	Thayammal Hospitals	11.6237472, 78.1177631	30 km/h	Medium
14	hospital	Vanitha Hospital	11.6239931, 78.1184622	30 km/h	Medium
16	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
17	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
18	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
19	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
20	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium
21	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
22	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
23	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
24	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
25	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
26	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
31	hospital	Jaaidev Hospital	11.6655205, 78.2634736	30 km/h	Medium
33	hospital	Niharika Hospital	11.6565971, 78.3937356	30 km/h	Medium
34	hospital	Malar Hospital	11.6567854, 78.3928345	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
8	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
10	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
11	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
12	school	Sri Vidhya Bharati Matric HR Sec School	11.6186069, 78.1103239	30 km/h	Medium
15	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
27	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium
28	school	Thangam ZEE public school	11.6653353, 78.251641	30 km/h	Medium
29	school	Sri Vidhya Mandir Senior Secondary School	11.668421, 78.2509573	30 km/h	Medium
30	college	AVS College od Arts and Science, Salem.	11.6642672, 78.2584134	30 km/h	Medium
32	college	Mahendra Engineering College	11.6672537, 78.2641303	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63266, 78.12699



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45358, 77.85781



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45853, 77.85139



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45906, 77.85132



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46316, 77.84921



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45636, 77.81678



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.62145, 78.10740



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.62130, 78.10776



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.61235, 78.10438



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.61219, 78.10450



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63018, 78.12472



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.63028, 78.12474



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.62986, 78.12425



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.63304, 78.12738



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.65458, 78.37988



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.65479, 78.37995



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65602, 78.38701
